

Chenxu Niu

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EDUCATION

Ph.D., Computer Science, Texas Tech University

Lubbock, TX

HPC, Sustainable AI, Inference Framework, Metadata Management

06/2019 – 12/2025

- Dissertation: "Accelerating Scientific Dataset Discovery with Semantic Metadata Search and Large Language Models"

M.S., Information Science, University of Science and Technology of China

Hefei, China

GPA: 3.95 out of 4.0

09/2015 – 06/2018

B.E., Mathematics, University of Science and Technology of China

Hefei, China

GPA: 3.65 out of 4.0

09/2011 – 06/2015

EXPERIENCE

Solutions Architect, HPC System Engineer

01/2026 – Now

NVIDIA

Santa Clara, CA

- Working with NVIDIA AI Native, Consumer Internet and Enterprise customers on large data center GPU server and networking system deployments as Solution Architect Engineer
- Guide customer discussions on network design, compute/storage and support bring up of server/network/cluster deployments.
- Analyze and debug compute/network configuration, performance issues to deliver performant clusters

HPC Systems Administrator & Research Assistant

06/2019 – 12/2025

Texas Tech University

Lubbock, TX

- **Served as the primary cluster administrator**, responsible for configuring, maintaining, and troubleshooting the SLURM workload manager and software stack to support hundreds of users
- Designed **real-time monitoring infrastructure** tracking power consumption, thermal management, and resource utilization across entire data center with iDrac, Redfish and Grafana
- **Automated deployment, configuration, and system updates** of cluster services using Ansible
- Installed, configured, and optimized **Finite Element Analysis software** and other engineering/scientific applications for NSF ACCESS users on HPC systems
- Deployed and optimized **distributed LLM training and inference** on GPU Clusters
- Accelerated semantic metadata search on scientific datasets with **RAG**

TECHNICAL SKILLS

Programming & DevOps: Python, C/C++, CUDA, Docker, Kubernetes, Terraform, Ansible, Git

HPC & Infrastructure: Warewulf, Slurm, Spack, InfiniBand, Lustre/GPFS, Linux Admin

AI/ML Frameworks: PyTorch, vLLM, TensorRT-LLM, Ray, DeepSpeed, Hugging Face, CUDA programming

Distributed Systems: MPI, OpenMP, Apache Spark, Redis, message queues, microservices architecture

Cloud & Monitoring: AWS/GCP, Prometheus, Grafana, ELK stack, power management APIs, thermal monitoring

Databases & Storage: PostgreSQL, MongoDB, HDF5, NetCDF, Distributed filesystems, object storage

KEY PROJECTS

HPC Data Center Design & Implementation | NSF REPACSS Project

09/2023 – 12/2025

- Led end-to-end design and deployment of renewable energy-powered HPC facility
- Implemented comprehensive monitoring stack (iDrac/Redfish/Grafana/TimeScaleDB) tracking real-time PUE, carbon footprint, and thermal management across 1000+ sensors
- Worked on a high-performance network topology with 200Gbps InfiniBand HDR fabric and 100Gbps Ethernet switching, optimizing MPI communication patterns for AI/ML workloads

Distributed LLM Inference Monitoring and Optimization | vLLM, TensorRT-LLM, Ray

09/2023 – 06/2025

- Built real-time monitoring dashboard tracking GPU utilization, memory bandwidth, power draw, and thermal throttling across distributed inference clusters
- Created automated benchmark suite comparing vLLM, TensorRT-LLM, and Ray Serve performance across different hardware configurations and model sizes
- Developed a benchmark named **TokenPowerBench** to measure the energy efficiency and performance of open-source large language models across different inference engines and deployment configurations

SELECTED PUBLICATIONS

1. H. Chen, X. Liu, Y. Liu, J. Jiang, X. Liu, **C. Niu** and B. He. **The 1/W Law: Context Length is the Dominant Energy Lever in LLM Inference Fleets**. Accepted to appear in *Advances in Neural Information Processing Systems (NeurIPS '26)*, 2026. (Proceedings forthcoming.)
2. Jia Xiong, Runkai Li, **Chenxu Niu**, Guangyuan Gao, Changwen Xing, Yifan Zhang, Xinlai Wan, Jieran Cui, Chen Bai, Yusheng Hua, Ying Wang, Ming Ling, Xi Wang and Tao Xie. **MicroEvo: Knowledge-Guided LLM Sampling for Efficient Microarchitecture Design Space Exploration**. Accepted to appear in *Proceedings of the 2026 IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26)*, 2026. (Proceedings forthcoming.)
3. **Chenxu Niu**, Wei Zhang, Jie Li, Yongjian Zhao, Tongyang Wang, Xi Wang and Yong Chen. **TokenPowerBench: Benchmarking the Power Consumption of LLM Inference**. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI '26)*, 2026. (Acceptance Rate: 17.6%)
4. Gwok-Waa Wan, SamZaak Wong, Shengchu Su, **Chenxu Niu**, Ning Wang, Xinlai Wan, Qixiang Chen, Mengnv Xing, Jingyi Zhang, Jianmin Ye, Yubo Wang, RongChang Song, Tao Ni, Qiang Xu, Nan Guan, Zhe Jiang, Xi Wang, Yong Chen, Jun Yang. **FIXME: Towards End-to-End Benchmarking of LLM-Aided Design Verification**. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence (AAAI '26)*, 2026. (Acceptance Rate: 17.6%)
5. Yongjian Zhao, Jie Li, **Chenxu Niu**, Alan Sill and Yong Chen. **Power-Centric Observability for HPC Systems: Design, Deployment, and Evaluation on REPACSS**. In *Proceedings of the Practice and Experience in Advanced Research Computing 2026 (PEARC '26)*, pp. 1–8, 2026.
6. **Chenxu Niu**, Wei Zhang, Yongjian Zhao and Yong Chen. **Energy Efficient or Exhaustive? Benchmarking Power Consumption of LLM Inference Engines**. In *ACM SIGENERGY Energy Informatics Review, Volume 5 Issue 2*, 2025.
7. **Chenxu Niu**, Wei Zhang, Mert Side and Yong Chen. **ICEAGE: Intelligent Contextual Exploration and Answer Generation Engine for Scientific Data Discovery**. In *Proceedings of the 37th ACM International Conference on Scalable Scientific Data Management (SSDBM'25)*, 2025.
8. **Chenxu Niu**, Wei Zhang, Suren Byna and Yong Chen. **PSQS: Parallel Semantic Querying Service for Self-describing File Formats**. In *Proceedings of the 13th IEEE International Conference on Big Data (BigData'23)*, 2023.
9. **Chenxu Niu**, Wei Zhang, Suren Byna and Yong Chen. **Kv2vec: A Distributed Representation Method for Key-value Pairs from Metadata Attributes**. In *Proceedings of the 27th IEEE High Performance Extreme Computing Conference (HPEC'22)*, 2022.
10. Wei Zhang, Suren Byna, **Chenxu Niu** and Yong Chen. **Exploring Metadata Search Essentials for Scientific Data Management**. In *Proceedings of 26th IEEE International Conference on High Performance Computing, Data, and Analytics (HiPC '19)*, 2019.

PROFESSIONAL SERVICE

Program Committee Member: **AAAI '27, AAAI '26, PEARC '26**
Reproducibility Committee Member: **SC '25**
Paper Reviewer: **SC '25, AAAI '26, BigData '25, PEARC '26**
Conference Volunteer: **SC '21 and SC '24**

TEACHING EXPERIENCE

Graduate Teaching Assistant	2020 – 2022
<i>Texas Tech University (TTU)</i>	<i>Lubbock, Texas</i>
• Computational Thinking with Data Science (Fall 2021 - Fall 2022) • Advanced Operating System Design (Spring 2021) • Analysis of Algorithms (Spring 2020)	

INVITED TALKS & PRESENTATIONS

Semantic Search and Natural Language Query over HDF5. *2024 HDF5 User Group Meeting (HUG24)*, 2024.